

Dept. of Computer and Communication Engineering

Patuakhali Science and Technology University

4<sup>th</sup> Semester (Level-2, Semester-II), Midterm Examination of B.Sc. Engg. (CSE), July-December: 2020

Course Code: CCE-223 Course Title: Database System

Credit Hour: 3.0 Full Marks: 15 Duration: 40 Minutes

- 1 a) Suppose you want to build a video site similar to Netflix. Consider each of the points listed in purpose of the databases as disadvantages of keeping data in a file-processing system. Discuss the relevance of each of these points to the storage of actual video data, and to metadata about the video, such as title, the user who uploaded it, tags, and which users viewed it. 4
- b) List five responsibilities of a database-management system. For each responsibility, explain the problems that would arise if the responsibility were not discharged. 4
- c) Sketch and explain a simplified database system environment. 3
- (d) Classify the types of database end users with example. Discuss the main activities of each. 4

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In force security.

In force constraints.

Register and break up.

concerning : Maintenance

- 3 [A.] Distinguish among SQL, MYSQL, Oracle and SQL Server. What is the difference between CHAR and VARCHAR? 3
- [B.] Explain different types of SQL constraints and integrity constraints. 3
- [C.] Draw the schema diagram of the PSTU university database. 5
- [D.] Consider above university database
- Find the titles of courses in the Comp. Sci. department that have 3 credits.
  - Find the IDs of all students who were taught by an instructor named Einstein; make sure there are no duplicates in the result.
  - Find the ID and name of each student who has taken at least one Comp. Sci. course; make sure there are no duplicate names in the result.
  - Find the course id, section id, and building for each section of a Biology course.
  - Output instructor names sorted by the ratio of their salary to their department's budget (in ascending order).
- 4 [A.] Explain the different types of attributes with an appropriate example. 7
- [B.] What is the significance of normalization in database design? Describe different types of normalization with appropriate example. 7
- 5 [A.] Define various types of keys with appropriate examples and distinguish them from one another. 7
- [B.] i. Why and in which case *Triggers* can be used? What system privileges are required to create a *trigger* on a table? 7
- ii. Specify the total and partial participation in the data base management system seen.
- 6 [A.] i. Define relationship and relationship set. 7
- ii. If you want to *Lock* after five consecutive failed connection attempts to the JANE account, what will have to do in *Oracle* for this?
- [B.] University wants to give scholarship to some students on the following criteria: 7
- Students must be female
  - Student do not get any other private scholarships such like DBBL (Dutch-Bangla Bank Ltd.) scholarship
  - Grade must be at least 3.50
  - Student should not be punished for any awful activity
- Create necessary table (yourself) and write necessary query for i, ii, iii and iv.

**Patuakhali Science and Technology University**  
**4<sup>th</sup> Semester Final Examination of B.Sc. Engineering in CSE Level: 2 Semester: II**

Course Code  
CCE 223

Course Title  
Database System

July-December  
2019

Credit: 03  
Time: 03 Hr  
Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged and write the full question number e.g. 3(A)(ii) before the answer paragraph)

- ✓ A What are advantages of DBMS over traditional file-based systems? What are super, primary, candidate and foreign keys?
- 1 B What is query processor? Describe the three-tier database architecture.
- 1 ✓ C Describe the differences in meaning between the term's *relation* and *relation schema*.
- 1 ✓ D Consider the following relational schema

employee(empno, name, office, age)  
 books(isbn, title, authors, publisher)  
 loan(empno, isbn, date)

Write the following queries in relational algebra.

- Find the names of employees who have borrowed a book published McGraw-Hill.
- Find the names of employees who have borrowed all books published McGraw-Hill.
- Find the names of employees who have borrowed more than five different books published McGraw-Hill.
- For each publisher, find the names of employees who have borrowed more than five books of that publisher.

2 ✓ A. Define DDL and DML. Write down the basic SQL domain type.

2 ✓ B. Consider the "Customers" table of a sample database:

| CustomerID | CustomerName                       | ContactName        | Address                       | City        | PostalCode | Country |
|------------|------------------------------------|--------------------|-------------------------------|-------------|------------|---------|
| 1          | Alfreds Futterkiste                | Maria Anders       | Obere Str. 57                 | Berlin      | 12209      | Germany |
| 2          | Ana Trujillo Emparedados y helados | Ana Trujillo       | Avda. de la Constitución 2222 | México D.F. | 05021      | Mexico  |
| 3          | Antonio Moreno Taquería            | Antonio Moreno     | Mataderos 2312                | México D.F. | 05023      | Mexico  |
| 4          | Around the Horn                    | Thomas Hardy       | 120 Hanover Sq.               | London      | WA1 1DP    | UK      |
| 5          | Berglunds snabbköp                 | Christina Berglund | Berguvsvägen                  | Luleå       | S-958 22   | Sweden  |

Write down the SQL statements for the following conditions

- Selects the "CustomerName" and "City" columns
- Selects ALL (including the duplicates) values from the "Country" column
- Selects all the customers from the country "Mexico"
- Select all records where the City column has the value "Berlin"
- Selects all fields from "Customers" where country is "Germany" AND city is "Berlin"
- Selects all fields from "Customers" where country is NOT "Germany"
- Selects all customers from the "Customers" table, sorted by the "Country" column
- Inserts a new record in the "Customers" table
- Lists all customers with a NULL value in the "Address" field
- Update the City column of all records in the Customers table
- Deletes the customer "Alfreds Futterkiste"

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3. A Explain the concept of ACID properties in DBMS? List two reasons why null values might be introduced into the database. 03

3 B Consider the Customer Table of 2(B) and Order Table of the following 08

| OrderID | CustomerID | EmployeeID | OrderDate  | ShipperID |
|---------|------------|------------|------------|-----------|
| 10308   | 2          | 7          | 1996-09-18 | 3         |
| 10309   | 37         | 3          | 1996-09-19 | 1         |
| 10310   | 77         | 8          | 1996-09-20 | 2         |

Write down the SQL statements for the following conditions

- selects all orders with customer and shipper information (inner join)
  - select all customers, and any orders they might have (left join)
  - selects all customers, and all orders (Full outer join)
  - selects matches customers that are from the same city (Self join)
- C What are the purposes of report generators? Discuss about web interfaces to databases. 03

- ✓ A UGC wants to give scholarship to some students on the following criteria: 07
- Students must be female
  - Student do not get any other private scholarships such like DBBL (Dutch-bangla bank Ltd.) scholarship
  - Grade must be at least 3.50
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- Create necessary table (yourself) and write necessary query for i, ii, iii and iv.

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4 B Define weak entity set. Describe weak entity set with a suitable example. 7

✓ A Describe each line of the following Trigger in Oracle. 7

```

i. create or replace trigger BOOKSHELF_BEF_UPD_ROW
ii. before update on BOOKSHELF
iii. for each row
iv. when (new.Rating < old.Rating)
v. begin
vi. insert into BOOKSHELF_AUDIT
vii. (Title, Publisher, CategoryName,
viii. Old_Rating, New_Rating, Audit_Date)
ix. values
x. (:old.Title, :old.Publisher, :old.CategoryName,
xi. :old.Rating, :new.Rating, Sysdate);
xii. end;
xiii. /

```

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=>=

5 B What is normalization? Describe different types of normalization with appropriate example. 7

- ✓ A 7
- What is Triggers? What are the types of Triggers?
  - If you want to Lock after four consecutive failed connection attempts to the 'MASUD' account, what will have to do in Oracle for this?

6 B What are the different types of attributes? Explain with suitable example. 7

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31 done SELECT \* FROM table